# ACTL3142 Tutorial

## Week 5: Generalised Linear Models


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# Recap: From Last Week

Last week (Week 4) we built two models for non-Normal responses. Both used the same linear predictor $\eta = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p$:

- **Logistic regression** (binary $Y$): logit link, so $$P(Y = 1) = \frac{1}{1 + e^{-\eta}}$$
- **Poisson regression** (count $Y$): log link, so $\mathrm{E}[Y] = e^{\eta}$

Both are fit by **maximum likelihood**, and both read off coefficients *multiplicatively*: a one-unit rise in $X_j$ scales the odds (or the mean) by $e^{\beta_j}$.

We ended Week 4 by noting these are not separate tricks. Logistic, Poisson, and even ordinary linear regression are all instances of *one* framework: a distribution + a link + a linear predictor.

Today: that framework in full, **Generalised Linear Models**, the exponential dispersion family, link functions, fitting, deviance, and diagnostics.

# Why Do We Need GLMs?

OLS assumes $Y \sim N(\mu, \sigma^2)$: a Normal response, *constant* variance, and a mean free to land anywhere on $(-\infty, \infty)$.

Insurance data rarely cooperates:

- **Claim counts**: non-negative integers $0, 1, 2, \ldots$ (Week 4: Poisson)
- **Claim occurrence**: binary 0/1 (Week 4: logistic)
- **Claim amounts**: positive and right-skewed (new: Gamma)

Forcing OLS here predicts negative counts or amounts, and the variance is wrong: the spread of claim sizes *grows* with their mean, it is not constant.

## What a GLM adds to a linear predictor

(1) A **response distribution** matched to the data (count, binary, positive-skewed), and (2) a **link** that keeps the mean in its valid range. Week 4 did this case by case; GLMs give the general recipe. With a log link, effects become *multiplicative*, $\mu = e^{\beta_0} e^{\beta_1 x_1} \cdots$, exactly how insurance rating factors stack.

# The Three Components of a GLM (Q2)

**Notation.** $\mathbf{X}$ is the $n \times (p+1)$ **design matrix** (one row per observation); $\mathbf{x}_i$ is its $i$-th row, the covariates for observation $i$ (with a leading 1 for the intercept). A GLM has three ingredients:

1. **Systematic component** (linear predictor):

$$\eta_i = \mathbf{x}_i \boldsymbol{\beta} = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip}$$

2. **Random component** (distribution): each response, given its own covariates, is in the exponential dispersion family

$$Y_i \mid \mathbf{x}_i \sim \mathrm{EDF}(\theta_i, \psi)$$

3. **Link function** connecting them:

$$g(\mu_i) = \eta_i, \qquad \mu_i := \mathrm{E}[Y_i \mid \mathbf{x}_i]$$

*We condition on $\mathbf{x}_i$, not the whole $\mathbf{X}$: observation $i$'s distribution is driven only by its own covariate row. OLS is the special case Normal + identity link $g(\mu) = \mu$.*

# What a GLM Really Models

A GLM gives **each observation its own distribution**, all drawn from the same EDF family:

$$Y_i \mid \mathbf{x}_i \sim \mathrm{EDF}(\theta_i, \psi), \qquad \mu_i = \mathrm{E}[Y_i \mid \mathbf{x}_i].$$

- Same family and same dispersion $\psi$ for every $i$; observations differ *only* through their mean $\mu_i$ (canonical parameter $\theta_i$)
- We model that mean, and only the mean: $g(\mu_i) = \eta_i = \mathbf{x}_i\boldsymbol{\beta}$

Each point has its own distribution (distributions vary on the canoncial parameter, $\theta_i$), but not its own free parameter

The $n$ means $\mu_1, \ldots, \mu_n$ are **not** $n$ free parameters: they are all generated by the *same* $p+1$ coefficients $\boldsymbol{\beta}$. Giving each its own free mean ($\hat{\mu}_i = y_i$) is the **saturated model** (it just memorises the data). A GLM is the disciplined middle ground: a few $\boldsymbol{\beta}$ explain all $n$ distributions, usually with $p << n$ of course.

# The Exponential Dispersion Family (EDF)

The stochastic component requires $Y$ to belong to the EDF. A distribution is in the EDF if its density can be written as:

$$f_Y(y; \theta, \psi) = \exp\left[\frac{y\theta - b(\theta)}{\psi} + c(y; \psi)\right]$$

- $\theta =$ **canonical parameter**: fixes the mean. This is what we let depend on the predictors (so $\theta_i$ varies across observations)
- $\psi =$ **dispersion parameter**: controls spread. Shared across observations, it does *not* depend on the predictors
- $b(\theta) =$ cumulant function (determines which distribution)
- $c(y; \psi) =$ normalising term (does not involve $\theta$)

*To show a distribution is in the EDF, take its density and rearrange into this form, reading off $\theta$, $b(\theta)$ and $\psi$. That's exactly what Q3 and Q4(a) ask you to do.*

# Key EDF Properties

From the EDF form, two results drop out:

**Mean:**

$$E[Y] = b'(\theta) = \mu$$

**Variance:**

$$\mathrm{Var}(Y) = \psi\, b''(\theta) = \psi\, V(\mu)$$

where $V(\mu) = b''(\theta)$ is the **variance function**.

Why this matters

The variance function $V(\mu)$ tells you how the variance relates to the mean. For Poisson, $V(\mu) = \mu$: higher mean $\Rightarrow$ higher variance. This is built into the model, unlike OLS which assumes constant variance.

# Common EDF Members

| Dist. | $\theta$ | $b(\theta)$ | $\psi$ | $V(\mu)$ | Canon. link |
|---|---|---|---|---|---|
| Normal | $\mu$ | $\theta^2/2$ | $\sigma^2$ | $1$ | Identity |
| Poisson | $\ln\mu$ | $e^\theta$ | $1$ | $\mu$ | Log |
| Gamma | $-1/\mu$ | $-\ln(-\theta)$ | $1/\nu$ | $\mu^2$ | Reciprocal |
| Inv. Gauss. | $\frac{-1}{2\mu^2}$ | $-\sqrt{-2\theta}$ | $1/\lambda$ | $\mu^3$ | $1/\mu^2$ |
| Binomial | $\ln\frac{p}{1-p}$ | $m\ln(1+e^\theta)$ | $1$ | $\mu(1-\mu/m)$ | Logit |

*You don't need to memorise this, but you do need to be able to derive these from the density (Q3, Q4a, Q6). The variance function column is the most useful for understanding each distribution's behaviour.*

# Choosing an EDF in Practice

You usually **cannot plot** $Y \mid \mathbf{x}_i$ directly: with several predictors, no two observations share the same $\mathbf{x}_i$. So we choose in stages:

- **Support of** $Y$ (its histogram) picks the family: integer $\geq 0 \rightarrow$ count; $0/1 \rightarrow$ binary; positive and skewed $\rightarrow$ Gamma / Inverse Gaussian
- **Mean–variance relationship** picks *within* the family: bin the data, plot sample variance against sample mean. The slope of $\log \mathrm{Var}$ vs $\log \mu$ estimates $p$ in $V(\mu) = \mu^p$ (0 Normal, 1 Poisson, 2 Gamma, 3 IG)
- **Residuals after fitting** confirm or refute: deviance residuals vs fitted should have constant spread; a funnel shape means the variance function is wrong

## Marginal $\neq$ conditional

A skewed *marginal* histogram of $Y$ does **not** mean the conditional $Y \mid \mathbf{x}_i$ is skewed: the predictors can create that spread. Use the marginal for the *support*, not the shape. Choosing an EDF is iterative: guess from support, fit, check residuals, revise.

# Q3: Inverse Gaussian

⋆ Solution Consider the Inverse Gaussian distribution, as parametrised on Wikipedia (other parametrisations exist), i.e., with density:

3. $$f(y) = \sqrt{\frac{\lambda}{2\pi y^3}} \exp\left(-\frac{\lambda(y-\mu)^2}{2\mu^2 y}\right), \quad y > 0.$$

a. Show that this is a member of the exponential dispersion family. Hence, identify $\psi$ and $b(\theta)$. **Hint:** you can rely on your knowledge that $\theta = -1/2\mu^2$ (as seen in the lecture).
b. Find the mean and variance of the Inverse Gaussian (as function of $\mu$ and $\lambda$). You should rely on the properties of the exponential dispersion family, rather than doing tedious calculations ;-).

# Q3: Showing Inverse Gaussian is in the EDF

Start from the IG density and expand the exponent:

$$f(y) = \sqrt{\frac{\lambda}{2\pi y^3}} \exp\left[-\frac{\lambda(y-\mu)^2}{2\mu^2 y}\right]$$

$$-\frac{\lambda(y^2 - 2y\mu + \mu^2)}{2\mu^2 y} = \underbrace{-\frac{\lambda y}{2\mu^2} + \frac{\lambda}{\mu}}_{\text{depends on } \mu} \quad \underbrace{-\frac{\lambda}{2y}}_{\text{goes into } c}$$

Now rewrite the $\mu$-dependent terms:

$$-\frac{\lambda y}{2\mu^2} + \frac{\lambda}{\mu} = \frac{1}{1/\lambda}\left[y\left(\frac{-1}{2\mu^2}\right) - \left(\frac{-1}{\mu}\right)\right] = \frac{y\theta - b(\theta)}{\psi}$$

So: $\theta = -1/(2\mu^2)$, $\quad \psi = 1/\lambda$, $\quad b(\theta) = -\sqrt{-2\theta}$.
Then $E[Y] = b'(\theta) = (-2\theta)^{-1/2} = \mu$ and
$\mathrm{Var}(Y) = \psi\, b''(\theta) = \mu^3/\lambda$.

# What Is the Link Function?

The link function $g$ is just a transformation that connects the mean $\mu_i$ to the linear predictor:

$$g(\mu_i) = \eta_i = \mathbf{x}_i\boldsymbol{\beta}$$

**The identity link** is $g(\mu) = \mu$, so $\mu_i = \mathbf{x}_i\boldsymbol{\beta}$ directly, no transformation at all. This is exactly what OLS does:

$$\mathrm{E}[Y_i \mid \mathbf{x}_i] = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip}$$

So OLS is literally a GLM with Normal distribution + identity link.

Why does this work for Normal? Because $\mu_i$ can be any real number, and $\eta_i = \mathbf{x}_i\boldsymbol{\beta}$ can be any real number, so there is no mismatch. But if $\mu_i$ must be **positive** (claims) or between **0 and 1** (probabilities), the identity link breaks down and we need a transformation.

# Choosing a Link Function

The key question: **what range can $\mu$ take?** Since $\eta_i = \mathbf{x}_i\boldsymbol{\beta} \in (-\infty, +\infty)$, the link must map the range of $\mu$ onto this.

| **Range of $\mu$** | **Link** | $g(\mu)$ | **Effect type** |
|---|---|---|---|
| $(-\infty, +\infty)$ | Identity | $\mu$ | Additive |
| $(0, +\infty)$ | Log | $\ln \mu$ | Multiplicative |
| $(0, 1)$ | Logit | $\ln \frac{\mu}{1-\mu}$ | Odds ratios |

**Practical guidelines:**

- Counts (Poisson): **log**, ensures $\hat{\mu} > 0$
- Amounts (Gamma): **log**, positive predictions + multiplicative rating factors
- Binary (Binomial): **logit**, keeps $\hat{p} \in (0, 1)$
- Normal: **identity**, no constraint needed

*The* **canonical link** *($g(\mu) = \theta$) simplifies the math but isn't always the best practical choice. For Gamma, the canonical link is $1/\mu$, but log is preferred in insurance.*

# Q1: Gamma GLM Fitted Values

1. ⋆ Solution An insurance company tested for claim sizes under two factors, i.e. **CAR**, the insurance group into which the car was placed, and **AGE**, the age of the policyholder (i.e. two-way contingency table). It was assumed that the claim size $y_i$ follows a gamma distribution, under a log-link function. Analysis of a set of data for which $n = 8$ provided the following output, in the SAS software:

| Observation | Claim size | CAR type | Age group | Pred | Xbeta | Resdev |
|---|---|---|---|---|---|---|
| 1 | 27 | 1 | 1 | 25.53 | 3.24 | 0.30 |
| 2 | 16 | 1 | 2 | 24.78 | 3.21 | -1.90 |
| 3 | 36 | 1 | 1 | | 3.41 | 1.03 |
| 4 | 45 | 1 | 2 | 38.09 | 3.64 | 1.11 |
| 5 | 38 | 2 | 1 | 40.85 | 3.71 | -0.46 |
| 6 | 27 | 2 | 2 | 36.97 | 3.61 | -1.73 |
| 7 | 14 | 2 | 1 | | 2.45 | 0.69 |
| 8 | 6 | 2 | 2 | 14.59 | 2.68 | -2.55 |

Calculate the fitted claim sizes missing in the table.

# Q1: Computing Fitted Values

**Setup:** 8 claim observations with factors CAR (3 levels) and AGE (2 levels). Gamma response with log link. SAS gives $\hat{\eta}_i$ (the "Xbeta" column) but some fitted values are missing.

**Method:**

1. Build the design matrix row $\mathbf{x}_i$ (intercept + dummies for factor levels)
2. Compute the linear predictor: $\hat{\eta}_i = \mathbf{x}_i\hat{\boldsymbol{\beta}}$
3. Apply the **inverse link**: $\hat{\mu}_i = g^{-1}(\hat{\eta}_i) = e^{\hat{\eta}_i}$

Here, SAS already gives us $\hat{\eta}_i$, so we just exponentiate:

$$\hat{\mu}_3 = e^{3.41} = 30.27, \qquad \hat{\mu}_7 = e^{2.45} = 11.59$$

*The log link makes effects multiplicative. Changing a factor level multiplies the expected claim by $e^{\hat{\beta}_j}$, regardless of the other factor levels.*

# Fitting GLMs: Maximum Likelihood

We find $\hat{\boldsymbol{\beta}}$ by maximising the log-likelihood:

$$\ell(\boldsymbol{\beta}) = \sum_{i=1}^{n} \left[ \frac{y_i\,\theta_i - b(\theta_i)}{\psi} + c(y_i;\psi) \right]$$

**Where is $\boldsymbol{\beta}$?** It's hiding inside $\theta_i$. Each $\theta_i$ depends on the mean $\mu_i$, which depends on the linear predictor $\eta_i = \mathbf{x}_i\boldsymbol{\beta}$:

$$\boldsymbol{\beta} \;\longrightarrow\; \eta_i = \mathbf{x}_i\boldsymbol{\beta} \;\longrightarrow\; \mu_i = g^{-1}(\eta_i) \;\longrightarrow\; \theta_i$$

**The process:** take $\partial\ell/\partial\beta_j$, set each to zero, and solve. In general, these equations have no closed-form solution.

## In practice

`glm(y ~ x1 + x2, family = Gamma(link="log"))` handles everything.

# Deviance: The GLM Version of RSS

First, some notation. $\ell(\hat{\boldsymbol{\mu}})$ is the log-likelihood of your fitted model, where $\hat{\boldsymbol{\mu}}$ is the vector of fitted values. Two benchmark models:

**Saturated model** (the "perfect model"): uses **one parameter per observation**, so $\hat{\mu}_i = y_i$ exactly. It memorises the data perfectly. Its log-likelihood $\ell(\mathbf{y})$ is the highest achievable score.

**Null model** (the simplest model): uses **only an intercept**, so $\hat{\mu}_i = \bar{y}$ for everyone. Same prediction for every observation.

**Deviance** measures how far your model is from the saturated model:

$$D = 2\psi\left[\ell(\mathbf{y}) - \ell(\hat{\boldsymbol{\mu}})\right]$$

$D \geq 0$, and smaller is better. For Normal with identity link, $D = \text{RSS}$.

# Using Deviance: Testing Nested Models

If a simpler model is nested inside a bigger model, the drop in deviance tells you whether the extra variables are worth it.

**Rule of thumb:**

$$\boxed{D_{\text{small}} - D_{\text{big}} \; > \; 2 \times (\text{extra parameters}) \quad \Longrightarrow \quad \text{keep them}}$$

For a formal test (when $\psi$ is known, e.g. Poisson): compare $D_{\text{small}} - D_{\text{big}}$ to $\chi^2_{\Delta\text{d.f.},\,1-\alpha}$.

For **non-nested** models, use AIC instead (lower is better).

*R reports both null and residual deviance in `summary(glm(...))`. The drop between them shows how much your predictors helped.*

# Q4: Poisson GLM for French Auto Insurance

*Pull this question up on your side. Here are the key hints:*

**(a)** Rewrite $f(y) = e^{-\mu}\mu^y/y!$ as one `exp(...)`
$\rightarrow \theta = \ln \mu, \quad b(\theta) = e^\theta, \quad \psi = 1$

**(b)** Poisson log-likelihood: $\ell = \sum[y_i \ln \mu_i - \mu_i - \ln(y_i!)]$
Plug in $\mu_i = y_i$ (saturated) and $\mu_i = \hat{\mu}_i$ (fitted), subtract
The $\ln(y_i!)$ cancels

**(c)** Apply rule of thumb to the SAS deviance table
SEX and DRIVER_AGE: significant
VEH_AGE and LOYALTY: not significant

**(d)** Overdispersion: we'll cover this in Applied Q2

# Pseudo-$R^2$

$$R^2_{\text{pseudo}} = 1 - \frac{D_M}{D_0}$$

Measures how much of the **null deviance** is explained by your model. Same idea as $R^2 = 1 - \text{RSS}/\text{TSS}$ in OLS.

- $R^2_{\text{pseudo}} = 0$: your model is no better than predicting $\bar{y}$ for everyone
- $R^2_{\text{pseudo}} = 1$: your model fits as well as the saturated model

## Don't over-interpret

Pseudo-$R^2$ values for count and binary data are typically much lower than what you'd see in OLS. A value of 0.2–0.4 can be a perfectly good model. Don't compare it directly with an OLS $R^2$.

# Residuals for GLMs

**Pearson residuals:**

$$r_i^P = \frac{y_i - \hat{\mu}_i}{\sqrt{V(\hat{\mu}_i)}}$$

Raw residual divided by the standard deviation implied by the variance function.

**Deviance residuals:**

$$r_i^D = \mathrm{sign}(y_i - \hat{\mu}_i)\sqrt{d_i}$$

where $d_i$ is observation $i$'s contribution to the total deviance.

*Deviance residuals tend to be more normally distributed, so they're the default for diagnostic plots. Plot them against fitted values to check for patterns, just like in OLS.*

# Overdispersion (Preview)

Poisson assumes $\mathrm{Var}(Y) = \mu$. If the empirical variance in the data is **much larger** than the mean, the Poisson model fails.

We'll see a concrete example in Applied Q2, where $\hat{\psi} \approx 55$ (the variance is $55\times$ what Poisson predicts). The fix: use **Negative Binomial** instead.

# Applied Q2: Overdispersion in Practice

Third-party insurance claims data (`claims` vs `accidents`):

| **Model** | **AIC** | **Resid. Dev.** | **Issue** |
|---|---|---|---|
| `lm()` | – | (RSS) | Non-Normal residuals |
| Poisson GLM | 9796 | 8567 | $\hat{\psi} = 55.4$ |
| Neg. Binomial | 2024 | 190 | Much better fit |

The Poisson model has $\hat{\psi} \approx 55$: the observed variance is $\sim 55\times$ the mean. This is extreme.

Switching to Negative Binomial: AIC drops from 9796 to 2024, residual deviance from 8567 to 190, and diagnostic plots improve dramatically.

*Quasi-Poisson gives the same $\hat{\beta}$ as Poisson, but inflates SEs by $\sqrt{55.4} \approx 7.4$. It doesn't produce an AIC though, so you can't use it for model comparison.*

# Thanks for coming! See you next week.